

Algebra First-Year Exam, January 2023, Part 2

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Prove that the following polynomials are irreducible in $\mathbb{Q}[x]$, stating clearly any theorems you wish to apply.
 - (a) $x^3 - x^2 + x + 5$.
 - (b) $x^6 - 300$.
 - (c) $x^4 - 4x^3 + 2x^2 + 21x - 7$.
2. Show that the ring $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain with respect to the norm $N(a + b\sqrt{-2}) = a^2 + 2b^2$. (You may regard this ring as a subring of the field of complex numbers, with $\sqrt{-2} = i\sqrt{2}$.)
3. Let R and S be commutative rings with 1 and $\phi : R \rightarrow S$ a ring homomorphism with $\phi(1) = 1$. Let I be an ideal of S and $\phi^{-1}(I) = \{r \in R \mid \phi(r) \in I\}$.
 - (a) Show that $\phi^{-1}(I)$ is an ideal of R .
 - (b) Show that if I is a prime ideal then so is $\phi^{-1}(I)$.
 - (c) If I is maximal, is $\phi^{-1}(I)$ necessarily maximal? Give a proof or counterexample.
4. Let R be a ring with 1 and M a (unital) left R -module.
 - (a) State what it means for M to satisfy the Ascending Chain Condition (i.e. for M to be Noetherian).
 - (b) Prove that if M satisfies the ACC then M is finitely generated.
 - (c) Give an example of a finitely generated R -module M that does not satisfy the ACC.
5. Use the rational canonical form to find one representative of each conjugacy class of elements of order 6 in $\text{GL}(4, \mathbb{Q})$.
6. Let $K \subset F \subset E$ be fields.
 - (a) If $|F : K| = n$ and $|E : F| = m$, where m and n are finite, show that $|E : K| = nm$.
 - (b) Prove that if $|E : K|$ is finite, then $|F : K|$ and $|E : F|$ are both finite.